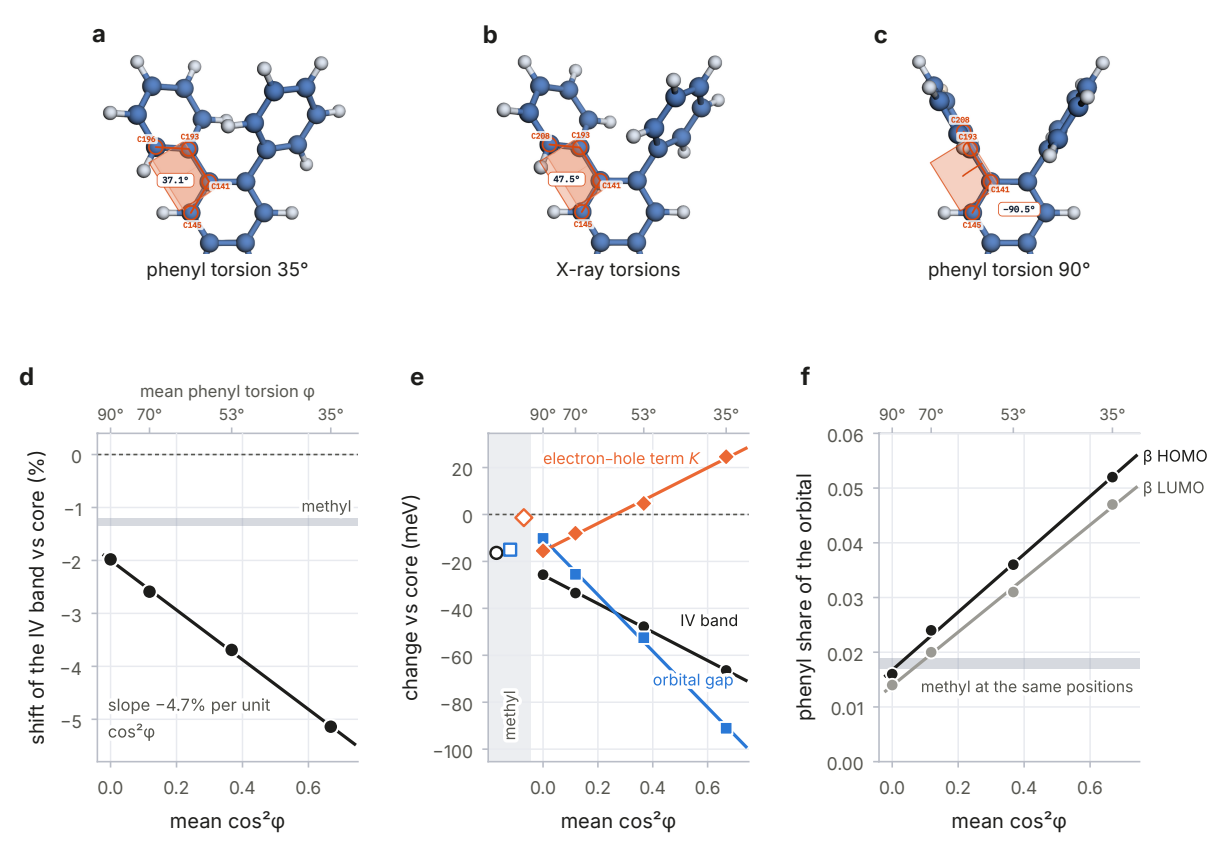

a
C196
C192
37.1°
C141
C145
phenyl torsion 35°
b
C208
C193
47.5°
C141
C145
X-ray torsions
c
C208
C193
C141
−90.5°
C145
phenyl torsion 90°
d
mean phenyl torsion φ
90° 70° 53° 35°
shift of the IV band vs core (%)
0
−1
−2
−3
−4
−5
methyl
slope −4.7% per unit
cos²φ
0.0 0.2 0.4 0.6
mean cos²φ
e
90° 70° 53° 35°
change vs core (meV)
20
0
−20
−40
−60
−80
−100
electron–hole term K
IV band
orbital gap
methyl
0.0 0.2 0.4 0.6
mean cos²φ
f
90° 70° 53° 35°
phenyl share of the orbital
0.06
0.05
0.04
0.03
0.02
0.01
0.00
β HOMO
β LUMO
methyl at the same positions
0.0 0.2 0.4 0.6
mean cos²φ